

ZIPCLIP: An AI-Powered Multi-Modal Platform for Automatic Video Summarization and Short-Form Content Generation

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ABSTRACT

The rapid growth of long-duration video content across digital platforms, including social media networks, online learning environments, and digital media repositories, has created an increasing demand for efficient methods to produce concise and engaging short-form videos. Short-form content has become widely preferred due to its ability to capture audience attention quickly and enhance content accessibility across diverse user groups. However, manually identifying and editing highlight segments from extended video material requires significant time, technical expertise, and sustained human effort, making the process inefficient and difficult to scale for large volumes of content. To address this limitation, this paper presents ZipClip, an intelligent video analysis framework designed to automatically extract meaningful short clips from lengthy video sources. The proposed system employs a multi-modal processing approach that integrates visual scene segmentation, audio-based speech and silence detection, emotion analysis, and semantic interpretation derived from textual transcripts. These complementary signals are systematically combined using a structured scoring mechanism to identify segments that are contextually relevant and emotionally significant while maintaining narrative coherence. By leveraging information from multiple modalities, the framework ensures balanced highlight selection without disrupting the logical flow of the original content.

General Terms

Artificial Intelligence, Multi-Modal Video Analysis, Video Summarization, Natural Language Processing, Short-Form Content Generation.

Keywords

AI-Based Video Summarization, Emotion Recognition, Highlight Detection, Multi-Modal Analysis, Short-Form Content Generation

1. INTRODUCTION

Digital platforms have witnessed a substantial rise in long-form video content, including lectures, podcasts, interviews, and panel discussions across domains such as education, entertainment, and social media. Although these videos often contain valuable and informative material, contemporary audiences increasingly prefer concise and engaging content that communicates key insights within a shorter duration. Consequently, content creators face growing pressure to transform lengthy recordings into compact, attention-focused segments optimized for short-form platforms. Traditionally, identifying meaningful moments within extended videos involves manual review and subjective editorial decision-making, a process that is time-intensive and difficult to scale for high-volume or frequent content production. Recent advancements in Artificial Intelligence (AI), particularly in machine learning, Natural Language Processing (NLP), and multi-modal audio-visual analysis, have enabled automated systems to interpret video data at a deeper semantic and contextual level. Leveraging these developments, ZipClip integrates scene boundary detection, speech and silence pattern analysis, emotion recognition, and transcript-based semantic understanding within a unified processing pipeline. By concurrently analyzing visual, audio, and textual modalities, the proposed framework systematically identifies segments that demonstrate contextual relevance and emotional significance while preserving narrative continuity. This multi-modal strategy reduces reliance on manual intervention and facilitates efficient repurposing of long-form content into structured short-form media suitable for modern digital distribution platforms.

2. LITERATURE SURVEY

2.1 AI-Based Video Summarization

Automated video summarization has emerged as a significant research area in response to the rapid expansion of long-duration video content within digital ecosystems. Existing approaches predominantly utilize deep learning models to capture temporal dependencies, visual saliency patterns, and semantic relevance across video sequences. These techniques aim to identify representative frames or segments that preserve essential information while reducing redundancy and overall duration. By modeling sequential relationships and visual importance, such systems have demonstrated improvements in minimizing manual editing effort. However, many existing methods primarily emphasize visual features, often under-representing contextual semantics and emotional dynamics that influence viewer engagement. This limitation highlights the need for more comprehensive multi-modal frameworks capable of integrating visual, auditory, and textual cues to enhance the quality and relevance of automatically generated summaries.[1-4].

2.2 Scene Change Detection and Structural Segmentation

Scene segmentation constitutes a fundamental component in automated video analysis, as it enables the division of continuous video streams into coherent and logically structured units. Techniques such as shot boundary detection, object-level change analysis, and learning-based segmentation approaches facilitate the identification of temporal transitions across diverse video formats. By partitioning videos into structurally consistent segments, these methods contribute to preserving narrative continuity and reducing abrupt visual discontinuities in automatically generated clips. Nevertheless, despite their effectiveness in detecting temporal boundaries, scene-based approaches alone do not sufficiently capture contextual importance or audience engagement factors, limiting their ability to determine which segments are most meaningful or impactful for viewers.[5-7].

2.3 Emotion Recognition in Video Analysis

Emotion-aware video analysis has emerged as an important approach for enhancing highlight selection by estimating potential audience engagement. By analyzing facial expressions, vocal intonation patterns, and contextual cues within speech, emotion recognition systems capture affective information that extends beyond conventional visual analysis. Recent multi-modal frameworks that integrate audio, visual, and textual features have demonstrated improved robustness and classification accuracy compared to uni-modal approaches. Such systems leverage complementary information across modalities to achieve a more comprehensive understanding of emotional dynamics within video content. However, despite these advancements, practical deployment remains challenging due to real-world factors including inconsistent lighting conditions, facial occlusions, background noise interference, and variability in speaker characteristics, which can significantly affect the reliability and stability of emotion estimation.[8-11].

2.4 Multi-Modal Fusion and Semantic Understanding

Recent research highlights the importance of multi-modal integration for achieving comprehensive video understanding. By combining visual structural information with speech content and transcript-based semantic analysis, automated systems can capture deeper contextual relationships within video segments. This in-

tegrated approach enables a more nuanced interpretation of content by aligning visual events with spoken dialogue and thematic context. Natural Language Processing (NLP) techniques further strengthen semantic analysis by identifying key topics, intent, and contextual relevance embedded within transcripts. Empirical studies indicate that multi-modal fusion strategies consistently outperform single-modality approaches, as they provide a more balanced and informed basis for highlight identification and content summarization.[4,9,11].

2.5 Challenges in Automated Video Clipping Systems

Despite notable advancements in automated video clipping systems, several limitations remain unresolved. Variations in video resolution, recording environments, lighting conditions, and audio clarity can significantly affect the reliability and consistency of automated analysis. Furthermore, fully automated highlight selection mechanisms may not always align with individual creator preferences, stylistic choices, or intended narrative emphasis. This discrepancy highlights the inherent challenge of balancing algorithmic decision-making with human creative control. Consequently, achieving an effective integration of intelligent automation and user-guided refinement remains a critical research objective, emphasizing the need for adaptable and context-aware video clipping frameworks.[8,10,12].

3. METHODOLOGY

3.1 Multi-Modal Structural Segmentation Models

The proposed framework adopts a structure-centric segmentation strategy to decompose long-form video content into coherent analytical units. Automated shot boundary detection and scene transition identification techniques are employed to capture temporal discontinuities and structural shifts within video streams. These segmentation mechanisms enable the system to establish logically consistent segments that preserve narrative continuity while reducing redundancy. Unlike purely visual summarization methods, structural segmentation in the proposed framework serves as a foundational layer that supports deeper contextual and affective analysis rather than acting as a standalone highlight selection criterion.

3.2 Semantic and Context-Aware Processing Approaches

To enhance contextual understanding, the system incorporates transcript-driven semantic analysis derived from automated speech recognition outputs. Natural Language Processing (NLP) techniques are applied to identify thematic relevance, keyword density, and contextual intent within each segmented unit. In parallel, silence detection and speech pattern analysis are utilized to identify emphasis regions and conversational dynamics. This combined audio-text processing enables a richer interpretation of informational importance beyond surface-level visual transitions. However, semantic interpretation alone may not sufficiently capture viewer engagement potential, necessitating integration with complementary analytical dimensions.

3.3 Emotion-Integrated Multi-Modal Analysis

The framework further integrates emotion-aware analysis by aggregating features extracted from visual cues, vocal characteristics, and transcript-derived affective indicators. Multi-modal emotion estimation strengthens robustness by compensating for limitations inherent in single-modality approaches. For instance, when facial

features are partially occluded or lighting conditions are inconsistent, vocal tonal variations and contextual sentiment cues contribute additional affective insight. This integrated strategy facilitates identification of segments demonstrating expressive intensity or heightened audience engagement potential, supporting more balanced highlight selection.

3.4 Implementation and System Integration Framework

From an implementation perspective, the framework is developed using modular backend services that support scalable AI inference and video processing operations. Python-based machine learning libraries, video processing tools such as OpenCV and FFmpeg, and NLP frameworks are utilized for feature extraction and analytical computation. RESTful APIs facilitate communication between processing modules, while centralized metadata storage maintains segment-level analytical outputs. Containerized deployment environments enhance portability and scalability, enabling efficient execution across diverse computational setups. This modular integration ensures extensibility while maintaining analytical consistency across system components.

4. RESULTS AND DISCUSSION

The evaluation of the ZipClip framework indicates that integrating multi-modal analysis improves the effectiveness of automated short-form video generation. By combining structural segmentation, speech and silence detection, emotion estimation, and transcript-based semantic analysis, the system is able to identify segments that demonstrate contextual relevance and expressive significance within long-form content. Observations across different video types, including lectures, podcasts, and interviews, suggest that the generated clips maintain narrative coherence while emphasizing informative and emotionally dynamic moments. Qualitative assessment further indicates that the inclusion of intent-aware refinement enhances adaptability by allowing creators to guide highlight emphasis through simple textual inputs. This feature supports a balance between automation and user influence, reducing manual editing effort while preserving creative flexibility. However, current evaluation primarily reflects functional validation rather than large-scale comparative benchmarking, and broader empirical studies are required to establish generalized performance metrics. During development and testing, practical challenges related to variability in video and audio quality were observed. Differences in lighting conditions, facial visibility, background noise, and speech variability affected the stability of emotion detection and transcript generation components. These findings highlight the sensitivity of multi-modal systems to real-world recording conditions and underscore the need for continued optimization to improve robustness, scalability, and long-term deployment reliability.

5. CONCLUSION

This paper presented ZipClip, a multi-modal AI framework for automated video summarization and short-form content generation. By integrating structural segmentation, speech and silence analysis, emotion-aware processing, and transcript-based semantic interpretation, the system addresses limitations associated with manual editing and single-modality highlight detection approaches. The proposed framework demonstrates the feasibility of combining complementary analytical components to support coherent and contextually meaningful clip extraction from long-form video content. The findings indicate that multi-modal integration contributes

to improved consistency in highlight identification while maintaining narrative continuity. Future work should focus on refining intent-driven personalization mechanisms, and conducting large-scale empirical evaluations using quantitative performance metrics. Addressing these areas will be essential for advancing automated video clipping systems toward more scalable, adaptable, and evidence-based deployment.

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